

# Zhengyuan Yang

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## Research Focus

I train frontier agentic models for long-horizon reasoning, tool use, and artifact generation, with coding as the shared foundation. My recent work on MAI-Code-1 centers on scaling reinforcement learning in executable environments, improving per-token intelligence, building reward and grading systems, and developing multimodal agent training recipes. Previously, I led diffusion distillation for DALL-E 3 and developed vision-language models and benchmarks for multimodal understanding and generation.

## Education

### University of Rochester

*Rochester, New York*

**PH.D. IN COMPUTER SCIENCE**     **ADVISOR:** PROF. JIEBO LUO

*2016 - 2021*

- Thesis: Visual Grounding: Building Cross-Modal Visual-Text Alignment (**ACM SIGMM Outstanding Ph.D. Thesis**)

### University of Science and Technology of China

*Hefei, China*

**B.E. IN ELECTRONIC ENGINEERING AND INFORMATION SCIENCE**

*2012 - 2016*

## Experience

### Microsoft

*Redmond, WA*

**MEMBER OF TECHNICAL STAFF, MAI SUPERINTELLIGENCE**

*June 2021 - Present*

- MAI-Code-1-Flash 2026
  - Token-efficient 5B-active coding model reached 71.6 on SBV with 10.8K tokens and 51.2 on SBP with 28.0K tokens.
  - Improved per-token intelligence via large-scale RL across 150K+ executable agentic environments with adaptive solution-length control, on 5B- and 35B-active models.
  - Built reward and grading systems, including aesthetic and functionality graders, for multimodal artifact generation across slides, graphics, games, UIs, and full-stack apps.
- Post-trained large-scale reasoning and agentic models for long-horizon tasks. 2025
  - Policy optimization in executable environments for SWE agents, DeepResearch, and other agentic tasks.
  - Improved GPT-OSS to 73.2% on SBV (+12%) and by 3% on the AAI Index using on-policy distillation and CoT RL.
  - Extended GPT-OSS to support vision inputs.
  - Post-trained multiple OpenAI and MAI models.
- Led diffusion distillation for DALL-E 3. 2024
  - Reduced diffusion from 64 steps to 1–4 while improving human-preference metrics.
  - Deployed the distilled models in Microsoft products and ChatGPT.
- Conducted research on scalable multimodal understanding and generation. 2023
  - Florence-GIT: scaled image-to-text transformers.
  - MM-Vet: a benchmark for large multimodal models.
  - “The Dawn of LLMs”: cataloged early GPT-4V use cases.

## Selected Publications

### MAI-Code-1-Flash

Microsoft AI Superintelligence team, “MAI-Code-1-Flash model card.”

### MAI-Thinking-1

Microsoft AI Superintelligence team, “MAI-Thinking-1: Building a Hill-Climbing Machine.”

### ICML 2026

Zihan Wang, Kangrui Wang, Qineng Wang, Pingyue Zhang, Linjie Li, **Zhengyuan Yang**, Xing Jin, Kefan Yu, Minh Nhat Nguyen, Licheng Liu, Eli Gottlieb, Yiping Lu, Kyunghyun Cho, Jiajun Wu, Li Fei-Fei, Lijuan Wang, Yejin Choi, Manling Li, “Understanding Reasoning Collapse in LLM Agent Reinforcement Learning,” The 43rd International Conference on Machine Learning (ICML), Seoul, South Korea, July 2026. **Spotlight presentation.**

### ACL 2026a

Cheng-Han Chiang, Xiaofei Wang, Linjie Li, Chung-Ching Lin, Kevin Lin, Shujie Liu, Zhendong Wang, **Zhengyuan Yang**, Hung-yi Lee, Lijuan Wang, “Shanks: Simultaneous Hearing and Thinking for Spoken Language Models,” The 64th Annual Meeting of the Association for Computational Linguistics (ACL), San Diego, CA, July 2026.

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| <b>ACL 2026b</b>     | Xiangxi Zheng, Linjie Li, <b>Zhengyuan Yang</b> , Ping Yu, Alex Jinpeng Wang, Rui Yan, Yuan Yao, Lijuan Wang, "V-MAGE: A Game Evaluation Framework for Assessing Vision-Centric Capabilities in Multimodal Large Language Models," The 64th Annual Meeting of the Association for Computational Linguistics (ACL), San Diego, CA, July 2026.                                    |
| <b>NeurIPS 2025a</b> | Xiyao Wang*, <b>Zhengyuan Yang*</b> , Chao Feng, Yongyuan Liang, Yuhang Zhou, Xiaoyu Liu, Ziyi Zang, Ming Li, Chung-Ching Lin, Kevin Lin, Linjie Li, Furong Huang, Lijuan Wang, "ViCrit: A Verifiable Reinforcement Learning Proxy Task for Visual Perception in VLMs," The 39th Conference on Neural Information Processing Systems (NeurIPS), San Diego, CA, December 2025.   |
| <b>NeurIPS 2025b</b> | Xiyao Wang, <b>Zhengyuan Yang</b> , Chao Feng, Hongjin Lu, Linjie Li, Chung-Ching Lin, Kevin Lin, Furong Huang, Lijuan Wang, "SoTA with Less: MCTS-Guided Sample Selection for Data-Efficient Visual Reasoning Self-Improvement," The 39th Conference on Neural Information Processing Systems (NeurIPS), San Diego, CA, December 2025. <b>Spotlight presentation.</b>          |
| <b>NeurIPS 2025c</b> | Minheng Ni, <b>Zhengyuan Yang</b> , Linjie Li, Chung-Ching Lin, Kevin Lin, Wangmeng Zuo, Lijuan Wang, "Point-RFT: Improving Multimodal Reasoning with Visually Grounded Reinforcement Finetuning," The 39th Conference on Neural Information Processing Systems (NeurIPS), San Diego, CA, December 2025.                                                                        |
| <b>NeurIPS 2025d</b> | Kangrui Wang, Pingyue Zhang, Zihan Wang, Yaning Gao, Linjie Li, Qineng Wang, Hanyang Chen, Yiping Lu, <b>Zhengyuan Yang</b> , Lijuan Wang, Ranjay Krishna, Jiajun Wu, Li Fei-Fei, Yejin Choi, Manling Li, "Reinforcing Visual State Reasoning for Multi-Turn VLM Agents," The 39th Conference on Neural Information Processing Systems (NeurIPS), San Diego, CA, December 2025. |
| <b>NeurIPS 2025e</b> | Jitesh Jain, <b>Zhengyuan Yang</b> , Humphrey Shi, Jianfeng Gao, Jianwei Yang, "Elevating Visual Perception in Multimodal LLMs with Visual Embedding Distillation," The 39th Conference on Neural Information Processing Systems (NeurIPS), San Diego, CA, December 2025.                                                                                                       |
| <b>EMNLP 2025</b>    | Yiyang Zhou, Linjie Li, Shi Qiu, <b>Zhengyuan Yang</b> , Yuyang Zhao, Siwei Han, Yangfan He, Kangqi Li, Haonian Ji, Zihao Zhao, Haibo Tong, Lijuan Wang, Huaxiu Yao, "GLIMPSE: LVLMs Can Watch but Do They Truly Understand?," The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP), Suzhou, China, Nov 2025.                                        |
| <b>ICCV 2025a</b>    | Xiyao Wang, <b>Zhengyuan Yang</b> , Linjie Li, Hongjin Lu, Yuancheng Xu, Chung-Ching Lin, Kevin Lin, Furong Huang, Lijuan Wang, "Scaling Inference-time Search with Vision Value Model for Improved Visual Comprehension," International Conference on Computer Vision (ICCV), Honolulu, Hawaii, Oct 2025.                                                                      |
| <b>ICCV 2025b</b>    | Jiaqi Liao, <b>Zhengyuan Yang</b> , Linjie Li, Dianqi Li, Kevin Lin, Yu Cheng, Lijuan Wang, "ImageGen-CoT: Enhancing Text-to-Image In-context Learning with Chain-of-Thought Reasoning," International Conference on Computer Vision (ICCV), Honolulu, Hawaii, Oct 2025.                                                                                                        |
| <b>ICCV 2025c</b>    | Wenqi Wang, Reuben Tan, Pengyue Zhu, Jianwei Yang, <b>Zhengyuan Yang</b> , Lijuan Wang, Andrey Kolobov, Jianfeng Gao, Boqing Gong, "SITE: towards Spatial Intelligence Thorough Evaluation," International Conference on Computer Vision (ICCV), Honolulu, Hawaii, Oct 2025.                                                                                                    |
| <b>ICML 2025a</b>    | Xingyu Fu, Minqian Liu, <b>Zhengyuan Yang</b> , John Corring, Yijuan Lu, Jianwei Yang, Dan Roth, Dinei Florencio, Cha Zhang, "ReFocus: Visual Editing as a Chain of Thought for Structured Image Understanding," The 42nd International Conference on Machine Learning (ICML), Vancouver, Canada, July 2025.                                                                    |
| <b>ICML 2025b</b>    | Yunzhuo Hao, Jiawei Gu, Huichen Will Wang, Linjie Li, <b>Zhengyuan Yang</b> , Lijuan Wang, Yu Cheng, "Can MLLMs Reason/Think in Multimodality? EMMA: An Enhanced MultiModal Reasoning Benchmark," The 42nd International Conference on Machine Learning (ICML), Vancouver, July 2025. <b>Oral presentation (1.0%).</b>                                                          |
| <b>CVPR 2025a</b>    | Kevin Qinghong Lin, Linjie Li, Difei Gao, <b>Zhengyuan Yang</b> , Shiwei Wu, Zechen Bai, Weixian Lei, Lijuan Wang, Mike Zheng Shou, "ShowUI: One Vision-Language-Action Model for GUI Visual Agent," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Nashville, TN, June 2025.                                                                               |
| <b>CVPR 2025b</b>    | Qin Liu, Jianfeng Wang, <b>Zhengyuan Yang</b> , Linjie Li, Kevin Lin, Marc Niethammer, Lijuan Wang, "LiVOS: Light Video Object Segmentation with Gated Linear Matching," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Nashville, TN, June 2025.                                                                                                           |
| <b>NAACL 2025</b>    | Chenglei Si, Yanzhe Zhang, Ryan Li, <b>Zhengyuan Yang</b> , Ruibo Liu, Diyi Yang, "Design2Code: Benchmarking Multimodal Code Generation for Automated Front-End Engineering," The Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL), New Mexico, April 2025. <b>Oral presentation.</b>                                                   |

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| <b>ICLR 2025a</b>    | Yining Hong, Beide Liu, Maxine Wu, Yuanhao Zhai, Kai-Wei Chang, Linjie Li, Kevin Lin, Chung-Ching Lin, Jianfeng Wang, <b>Zhengyuan Yang*</b> , Yingnian Wu*, Lijuan Wang*, "SlowFast-VGen: Slow-Fast Learning for Action-Driven Long Video Generation," The Thirteenth International Conference on Learning Representations (ICLR), Singapore, April 2025. <b>Spotlight presentation.</b> |
| <b>ICLR 2025b</b>    | Zichen Miao, <b>Zhengyuan Yang</b> , Kevin Lin, Ze Wang, Zicheng Liu, Lijuan Wang, Qiang Qiu, "Tuning Timestep-Distilled Diffusion Model Using Pairwise Sample Optimization," The Thirteenth International Conference on Learning Representations (ICLR), Singapore, April 2025.                                                                                                          |
| <b>ICLR 2025c</b>    | Yuyang Zhao, Chung-Ching Lin, Kevin Lin, Zhiwen Yan, Linjie Li, <b>Zhengyuan Yang</b> , Jianfeng Wang, Gim Hee Lee, Lijuan Wang, "GenXD: Generating Any 3D and 4D Scenes," The Thirteenth International Conference on Learning Representations (ICLR), Singapore, April 2025.                                                                                                             |
| <b>ICLR 2025d</b>    | Xuehai He, Weixi Feng, Kaizhi Zheng, Yujie Lu, Wanrong Zhu, Jiachen Li, Yue Fan, Jianfeng Wang, Linjie Li, <b>Zhengyuan Yang</b> , Kevin Lin, William Yang Wang, Lijuan Wang, Xin Eric Wang, "MMWorld: Towards Multi-discipline Multi-faceted World Model Evaluation in Videos," The Thirteenth International Conference on Learning Representations (ICLR), Singapore, April 2025.       |
| <b>ICLR 2025e</b>    | Kaizhi Zheng, Xiaotong Chen, Xuehai He, Jing Gu, Linjie Li, <b>Zhengyuan Yang</b> , Kevin Lin, Jianfeng Wang, Lijuan Wang, Xin Eric Wang, "EditRoom: LLM-parameterized Graph Diffusion for Composable 3D Room Layout Editing," The Thirteenth International Conference on Learning Representations (ICLR), Singapore, April 2025.                                                         |
| <b>NeurIPS 2024a</b> | Yuanhao Zhai, Kevin Lin, <b>Zhengyuan Yang</b> , Linjie Li, Jianfeng Wang, Chung-Ching Lin, David Doermann, Junsong Yuan, Lijuan Wang, "Motion Consistency Model: Accelerating Video Diffusion with Disentangled Motion-Appearance Distillation," The 38th Conference on Neural Information Processing Systems (NeurIPS), Vancouver, BC, December 2024.                                   |
| <b>NeurIPS 2024b</b> | Xueyan Zou, Linjie Li, Jianfeng Wang, Jianwei Yang, Mingyu Ding, Junyi Wei, <b>Zhengyuan Yang</b> , Feng Li, Hao Zhang, Shilong Liu, Arul Aravinthan, Yong Jae Lee, Lijuan Wang, "Interfacing Foundation Models' Embeddings," The 38th Conference on Neural Information Processing Systems (NeurIPS), Vancouver, BC, December 2024.                                                       |
| <b>NeurIPS 2024c</b> | Kevin Qinghong Lin, Linjie Li, Difei Gao, Qinchun Wu, Mingyi Yan, <b>Zhengyuan Yang</b> , Lijuan Wang, Mike Zheng Shou, "VideoGUI: A Benchmark for GUI Automation from Instructional Videos," The 38th Conference on Neural Information Processing Systems Datasets and Benchmarks (NeurIPS), Vancouver, BC, December 2024. <b>Spotlight presentation.</b>                                |
| <b>COLM 2024</b>     | An Yan, <b>Zhengyuan Yang</b> , Junda Wu, Wanrong Zhu, Jianwei Yang, Linjie Li, Kevin Lin, Jianfeng Wang, Julian McAuley, Jianfeng Gao, Lijuan Wang, "List Items One by One: A New Data Source and Learning Paradigm for Multimodal LLMs," The 1st Conference on Language Modeling (COLM), Philadelphia, PA, October 2024.                                                                |
| <b>ACMMM 2024</b>    | Jie An, <b>Zhengyuan Yang</b> , Linjie Li, Jianfeng Wang, Kevin Lin, Zicheng Liu, Lijuan Wang, Jiebo Luo, "Openleaf: Open-domain interleaved image-text generation and evaluation," ACM Multimedia Conference (MM), Brave New Ideas track, Melbourne, Australia, October 2024. <b>Oral presentation.</b>                                                                                  |
| <b>ECCV 2024a</b>    | <b>Zhengyuan Yang</b> , Jianfeng Wang, Linjie Li, Kevin Lin, Chung-Ching Lin, Zicheng Liu, Lijuan Wang, "Idea2Img: Iterative Self-Refinement with GPT-4V(ision) for Automatic Image Design and Generation," The 18th European Conference on Computer Vision (ECCV), Milano, Italy, Sept 2024.                                                                                             |
| <b>ECCV 2024b</b>    | Jialian Wu, Jianfeng Wang, <b>Zhengyuan Yang</b> , Zhe Gan, Zicheng Liu, Junsong Yuan, Lijuan Wang, "GRIT: A Generative Region-to-text Transformer for Object Understanding," The 18th European Conference on Computer Vision (ECCV), Milano, Italy, Sept 2024.                                                                                                                           |
| <b>ECCV 2024c</b>    | Yuanhao Zhai, Kevin Lin, Linjie Li, Chung-Ching Lin, Jianfeng Wang, <b>Zhengyuan Yang</b> , David Doermann, Junsong Yuan, Zicheng Liu, Lijuan Wang, "IDOL: Unified Dual-Modal Latent Diffusion for Human-Centric Joint Video-Depth Generation," The 18th European Conference on Computer Vision (ECCV), Milano, Italy, Sept 2024.                                                         |
| <b>ICML 2024a</b>    | Weihao Yu*, <b>Zhengyuan Yang*</b> , Linjie Li, Jianfeng Wang, Kevin Lin, Zicheng Liu, Xinchao Wang, Lijuan Wang, "MM-Vet: Evaluating Large Multimodal Models for Integrated Capabilities," The Forty-first International Conference on Machine Learning (ICML), Vienna, Austria, July 2024.                                                                                              |
| <b>ICML 2024b</b>    | Zecheng Tang, Chenfei Wu, Zekai Zhang, Mingheng Ni, Shengming Yin, Yu Liu, <b>Zhengyuan Yang</b> , Lijuan Wang, Zicheng Liu, Juntao Li, Nan Duan, "StrokeNUWA: Tokenizing Strokes for Vector Graphic Synthesis," The Forty-first International Conference on Machine Learning (ICML), Vienna, Austria, July 2024.                                                                         |

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| <b>IJCAI 2024</b>  | Jie An, <b>Zhengyuan Yang</b> , Jianfeng Wang, Linjie Li, Zicheng Liu, Lijuan Wang, Jiebo Luo, "Bring Metric Functions into Diffusion Models," The 32nd International Joint Conference on Artificial Intelligence (IJCAI), Jeju, August 2024.                                                                                                                                                                             |
| <b>CVPR 2024a</b>  | Chaoyi Zhang, Kevin Lin, <b>Zhengyuan Yang</b> , Jianfeng Wang, Linjie Li, Chung-Ching Lin, Zicheng Liu, Lijuan Wang, "MM-Narrator: Narrating Long-form Videos with Multimodal In-Context Learning," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, June 2024. <b>Highlight presentation.</b>                                                                                            |
| <b>CVPR 2024b</b>  | Tan Wang, Linjie Li, Kevin Lin, Yuanhao Zhai, Chung-Ching Lin, <b>Zhengyuan Yang</b> , Hanwang Zhang, Zicheng Liu, Lijuan Wang, "DisCo: Disentangled Control for Realistic Human Dance Generation," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, June 2024.                                                                                                                            |
| <b>CVPR 2024c</b>  | Zichen Miao, Jiang Wang, Ze Wang, <b>Zhengyuan Yang</b> , Lijuan Wang, Qiang Qiu, Zicheng Liu, "Training Diffusion Models Towards Diverse Image Generation with Reinforcement Learning," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, June 2024.                                                                                                                                       |
| <b>CVPR 2024d</b>  | Jielin Qiu, Jiacheng Zhu, William Han, Aditesh Kumar, Karthik Mittal, Claire Jin, <b>Zhengyuan Yang</b> , Linjie Li, Jianfeng Wang, Ding Zhao, Bo Li, Lijuan Wang, "MMSum: A Dataset for Multimodal Summarization and Thumbnail Generation of Videos," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, June 2024. <b>Highlight presentation.</b>                                          |
| <b>Book 2023</b>   | Chunyuan Li*, Zhe Gan*, <b>Zhengyuan Yang*</b> , Jianwei Yang*, Linjie Li*, Lijuan Wang, Jianfeng Gao, "Multimodal Foundation Models: From Specialists to General-Purpose Assistants," Foundations and Trends in Computer Graphics and Vision, 2023.                                                                                                                                                                      |
| <b>arXiv 2023a</b> | <b>Zhengyuan Yang*</b> , Linjie Li*, Kevin Lin*, Jianfeng Wang*, Chung-Ching Lin*, Zicheng Liu, Lijuan Wang*, "The Dawn of LMMs: Preliminary Explorations with GPT-4V(ision)." ( <b>Exploratory work cataloguing use of GPT-4V</b> ).                                                                                                                                                                                     |
| <b>arXiv 2023b</b> | <b>Zhengyuan Yang</b> , Linjie Li, Jianfeng Wang, Kevin Lin, Ehsan Azarnasab, Faisal Ahmed, Zicheng Liu, Ce Liu, Michael Zeng, Lijuan Wang, "MM-REACT: Prompting ChatGPT for Multimodal Reasoning and Action."                                                                                                                                                                                                            |
| <b>ICCV 2023a</b>  | Yushi Hu, Hang Hua, <b>Zhengyuan Yang</b> , Weijia Shi, Noah A. Smith, Jiebo Luo, "PromptCap: Prompt-Guided Task-Aware Image Captioning," IEEE International Conference on Computer Vision (ICCV), Paris, France, October 2023.                                                                                                                                                                                           |
| <b>ICCV 2023b</b>  | Tan Wang, Kevin Lin, Linjie Li, Chung-Ching Lin, <b>Zhengyuan Yang</b> , Hanwang Zhang, Zicheng Liu, Lijuan Wang, "Equivariant Similarity for Vision-Language Foundation Models," IEEE International Conference on Computer Vision (ICCV), Paris, France, October 2023. <b>Oral presentation.</b>                                                                                                                         |
| <b>ACL 2023</b>    | Shengming Yin, Chenfei Wu, Huan Yang, Jianfeng Wang, Xiaodong Wang, Minheng Ni, <b>Zhengyuan Yang</b> , Linjie Li, Shuguang Liu, Fan Yang, Jianlong Fu, Gong Ming, Lijuan Wang, Zicheng Liu, Houqiang Li, Nan Duan, "NUWA-XL: Diffusion over Diffusion for eXtremely Long Video Generation." Annual Meeting of the Association for Computational Linguistics (ACL), Toronto, Canada, July 2023. <b>Oral presentation.</b> |
| <b>IJCAI 2023</b>  | Xiaodong Wang, Chenfei Wu, Shengming Yin, Minheng Ni, Jianfeng Wang, Linjie Li, <b>Zhengyuan Yang</b> , Fan Yang, Lijuan Wang, Zicheng Liu, Yuejian Fang, Nan Duan, "Learning 3D Photography Videos via Self-supervised Diffusion on Single Images." The 32nd International Joint Conference on Artificial Intelligence (IJCAI), Macao, August 2023.                                                                      |
| <b>CVPR 2023</b>   | <b>Zhengyuan Yang</b> , Jianfeng Wang, Zhe Gan, Linjie Li, Kevin Lin, Chenfei Wu, Nan Duan, Zicheng Liu, Ce Liu, Michael Zeng, Lijuan Wang, "ReCo: Region-Controlled Text-to-Image Generation," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Vancouver, BC, June 2023.                                                                                                                              |
| <b>ICLR 2023</b>   | Chenglei Si, Zhe Gan, <b>Zhengyuan Yang</b> , Shuohang Wang, Jianfeng Wang, Jordan Boyd-Graber, Lijuan Wang, "Prompting GPT-3 To Be Reliable," The Eleventh International Conference on Learning Representations (ICLR), Kigali, Rwanda, May 2023.                                                                                                                                                                        |
| <b>ECCV 2022</b>   | <b>Zhengyuan Yang</b> , Zhe Gan, Jianfeng Wang, Xiaowei Hu, Faisal Ahmed, Zicheng Liu, Yumao Lu, Lijuan Wang, "UniTAB: Unifying Text and Box Outputs for Grounded Vision-Language Modeling," European Conference on Computer Vision (ECCV), Tel Aviv, Israel, October 2022. <b>Oral presentation (2.7%).</b>                                                                                                              |
| <b>TPAMI 2023</b>  | Jiajun Deng, <b>Zhengyuan Yang</b> , Daqing Liu, Tianlang Chen, Wengang Zhou, Yanyong Zhang, Houqiang Li, Wanli Ouyang, "TransVG++: End-to-End Visual Grounding with Language Conditioned Vision Transformer," IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2023.                                                                                                                              |

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| <b>TMLR 2022</b>   | Jianfeng Wang, <b>Zhengyuan Yang</b> , Xiaowei Hu, Linjie Li, Kevin Lin, Zhe Gan, Zicheng Liu, Ce Liu, Lijuan Wang, "GIT: A Generative Image-to-text Transformer for Vision and Language," Transactions on Machine Learning Research (TMLR), 2022.                                                                 |
| <b>CVPR 2022</b>   | Xiaowei Hu, Zhe Gan, Jianfeng Wang, <b>Zhengyuan Yang</b> , Zicheng Liu, Yumao Lu and Lijuan Wang, "Scaling Up Vision-Language Pre-training for Image Captioning," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), New Orleans, June 2022.                                                      |
| <b>arXiv 2021</b>  | Jianfeng Wang, Xiaowei Hu, Zhe Gan, <b>Zhengyuan Yang</b> , Xiyang Dai, Zicheng Liu, Yumao Lu and Lijuan Wang, "UFO: A UniFied TransFormer for Vision-Language Representation Learning."                                                                                                                           |
| <b>AAAI 2022</b>   | <b>Zhengyuan Yang</b> , Zhe Gan, Jianfeng Wang, Xiaowei Hu, Yumao Lu, Zicheng Liu, Lijuan Wang, "An Empirical Study of GPT-3 for Few-Shot Knowledge-Based VQA," The 36th AAAI Conference on Artificial Intelligence (AAAI), Vancouver, BC, February 2022. <b>Oral presentation (4.2%).</b>                         |
| <b>ICPR 2022</b>   | <b>Zhengyuan Yang</b> , Jingen Liu, Jing Huang, Xiaodong He, Tao Mei, Chenliang Xu, Jiebo Luo, "Cross-modal Contrastive Distillation for Instructional Activity Anticipation," International Conference on Pattern Recognition (ICPR), Montreal, Quebec, Canada, August 2022.                                      |
| <b>ICCV 2021a</b>  | <b>Zhengyuan Yang</b> , Songyang Zhang, Liwei Wang, Jiebo Luo, "SAT: 2D Semantics Assisted Training for 3D Visual Grounding," IEEE International Conference on Computer Vision (ICCV), Online, 2021. <b>Oral presentation (3.4%).</b>                                                                              |
| <b>ICCV 2021b</b>  | Jiajun Deng, <b>Zhengyuan Yang</b> , Tianlang Chen, Wengang Zhou, Houqiang Li, "TransVG: End-to-End Visual Grounding with Transformers," IEEE International Conference on Computer Vision (ICCV), Online, 2021.                                                                                                    |
| <b>CVPR 2021a</b>  | <b>Zhengyuan Yang</b> , Yijuan Lu, Jianfeng Wang, Xi Yin, Dinei Florencio, Lijuan Wang, Cha Zhang, Lei Zhang, Jiebo Luo, "TAP: Text-Aware Pre-training for Text-VQA and Text-Caption," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Online, June 2021. <b>Oral presentation (4.0%).</b>      |
| <b>CVPR 2021b</b>  | Liwei Wang, Jing Huang, Yin Li, Kun Xu, <b>Zhengyuan Yang</b> , Dong Yu, "Improving Weakly Supervised Visual Grounding by Contrastive Knowledge Distillation," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Online, June 2021.                                                               |
| <b>ECCV 2020</b>   | <b>Zhengyuan Yang</b> , Tianlang Chen, Liwei Wang, Jiebo Luo, "Improving One-stage Visual Grounding by Recursive Sub-query Construction," European Conference on Computer Vision (ECCV), Online, August 2020.                                                                                                      |
| <b>ACL 2020</b>    | Yongjing Yin, Fandong Meng, Jinsong Su, Chulun Zhou, <b>Zhengyuan Yang</b> , Jie Zhou, Jiebo Luo, "A Novel Graph-based Multi-modal Fusion Encoder for Neural Machine Translation," Annual Meeting of the Association for Computational Linguistics (ACL), Online, July 2020.                                       |
| <b>T-CSVT 2020</b> | <b>Zhengyuan Yang</b> , Tushar Kumar, Tianlang Chen, Jingsong Su, Jiebo Luo, "Grounding-Tracking-Integration," IEEE Transactions on Circuits and Systems for Video Technology (T-CSVT).                                                                                                                            |
| <b>ICCV 2019</b>   | <b>Zhengyuan Yang</b> , Boqing Gong, Liwei Wang, Wenbing Huang, Dong Yu, Jiebo Luo, "A Fast and Accurate One-Stage Approach to Visual Grounding," IEEE International Conference on Computer Vision (ICCV), Seoul, South Korea, 2019. <b>Oral presentation (4.3%).</b>                                              |
| <b>CVPR 2019</b>   | Mengshi Qi, Weijian Li, <b>Zhengyuan Yang</b> , Yunhong Wang, Jiebo Luo, "Attentive Relational Networks for Mapping Images to Scene Graphs," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Long Beach, USA, 2019.                                                                             |
| <b>T-CSVT 2018</b> | <b>Zhengyuan Yang</b> , Yuncheng Li, Jianchao Yang, Jiebo Luo, "Action Recognition with Spatio-Temporal Visual Attention on Skeleton Image Sequences," IEEE Transactions on Circuits and Systems for Video Technology (T-CSVT).                                                                                    |
| <b>ICPR 2018a</b>  | <b>Zhengyuan Yang</b> , Yixuan Zhang, Jerry Yu, Junjie Cai, Jiebo Luo, "End-to-end Multi-Modal Multi-Task Vehicle Control for Self-Driving Cars with Visual Perceptions," International Conference on Pattern Recognition (ICPR), Beijing, China, 2018. <b>Best Industry Related Paper Award (BIRPA) (1/1258).</b> |

## Talks

**Invited Talk – Data, Perception, and Agents: Post-Training for Long-Horizon Multimodal Reasoning**

*Champaign, IL*

• UIUC Computer Vision Group

2025

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|-----------------------------------------------------------------------------------------|------------------------------|
| <b>Tutorial Talk – Multimodal Reasoning for Visual-Centric Long-Horizon Tasks</b>       | <i>Nashville, TN</i>         |
| • CVPR 2025 Tutorial on Recent Advances in Vision Foundation Models                     | 2025                         |
| <b>Invited Talk – Benchmarks in the Era of Multimodal Reasoning and Agentic Models</b>  | <i>Nashville, TN</i>         |
| • CVPR 2025 Workshop on Computer Vision in the Wild                                     | 2025                         |
| <b>Invited Talk – Multimodal Agents</b>                                                 | <i>Niagara Falls, Canada</i> |
| • ICME 2024 Workshop on Visual-Language Alignment in Text-Guided Multi-Modal Generation | 2024                         |
| <b>Tutorial Talk – Recent Advances in Image Generative Foundation Models</b>            | <i>Seattle, WA</i>           |
| • CVPR 2024 Tutorial on Recent Advances in Vision Foundation Models                     | 2024                         |
| <b>Invited Talk – Multimodal Agents: from Text to Multimodal Reasoning and Action</b>   | <i>Hong Kong, China</i>      |
| • CUHK Multi-Modal Symposium                                                            | 2024                         |
| <b>Tutorial Talk – Alignments in Text-to-Image Generation</b>                           | <i>Vancouver, BC</i>         |
| • CVPR 2023 Tutorial on Recent Advances in Vision Foundation Models                     | 2023                         |
| <b>Invited Talk – Towards Cross-Modal Visual-Text Understanding and Generation</b>      | <i>Tokyo, Japan</i>          |
| • ACM MM Asia 2022 Workshop on Multimedia Understanding with Pre-trained Models         | 2022                         |
| <b>Tutorial Talk – Unified Image-Text Modeling</b>                                      | <i>New Orleans, LA</i>       |
| • CVPR 2022 Tutorial on Recent Advances in Vision-and-Language Pre-training             | 2022                         |
| <b>Invited Talk – SAT: 2D Semantics Assisted Training for 3D Visual Grounding</b>       | <i>Online</i>                |
| • CVPR Workshop 2021 on Language for 3D Scenes                                          | 2021                         |
| <b>Guest Lecture – Vision-and-language</b>                                              | <i>Rochester, NY</i>         |
| • CS440 Data Mining, University of Rochester                                            | 2021                         |
| <b>Guest Lecture – CNN and Feature Visualization</b>                                    | <i>Rochester, NY</i>         |
| • CS440 Data Mining, University of Rochester                                            | 2020                         |

## Selected Awards

|                                                                  |      |
|------------------------------------------------------------------|------|
| ACM SIGMM Award for Outstanding Ph.D. Thesis                     | 2022 |
| ECCV 2022 Outstanding Reviewer                                   | 2022 |
| Winner of CVPR 2021 TextCaps Challenge                           | 2021 |
| Winner of CVPR 2021 ReferIt3D Challenge                          | 2021 |
| CVPR 2021 Outstanding Reviewer                                   | 2021 |
| Twitch Research Fellowship                                       | 2020 |
| Best Industry Related Paper Award (BIRPA) in ICPR 2018, (1/1258) | 2018 |

## Services

|                                  |                                                                                                   |           |
|----------------------------------|---------------------------------------------------------------------------------------------------|-----------|
| <b>Exhibits and Demos Chair:</b> | IEEE International Conference on Multimedia and Expo (ICME)                                       | 2024      |
| <b>Area Chair:</b>               | ICLR, ICML, NeurIPS, EMNLP, ACL, NAACL, ACMMM                                                     |           |
| <b>Senior Program Committee:</b> | AAAI Conference on Artificial Intelligence (AAAI)                                                 | 2023-2026 |
| <b>Associate Editor:</b>         | IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)                            | 2022-2026 |
| <b>Guest Editor:</b>             | IEEE TMM SI on “Large Multimodal Models for Dynamic Visual Scene Understanding”                   | 2024      |
|                                  | IEEE TCSVT SI on “AI-Generated Content for Multimedia”                                            | 2023      |
| <b>Conference Reviewer:</b>      | CVPR, ICCV, ECCV, NeurIPS, ICLR, ICML, ACL, EMNLP, ACMMM, AAAI, ACCV, WACV, ICME, ICIP.           |           |
| <b>Journal Reviewer:</b>         | TPAMI, IJCV, TIP, TMM, TCybernetics, TCSVT, Pattern Recognition, Neurocomputing, TBioCAS, Access. |           |
| <b>Tutorial Organizer:</b>       | Recent Advances in Vision Foundation Models at CVPR                                               | 2022-2025 |
| <b>Tutorial Organizer:</b>       | Computer Vision in the Wild at CVPR                                                               | 2025      |